

Aditya Raut

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Work Experience

Tata Consultancy Services (TCS), System Engineer

Apr 2021 - Oct 2023

Clients: GE Aerospace, RBI

- Architected and developed backend services for the **Centralized Information Management System (CIMS)**, a large-scale data warehouse and data lake platform ingesting and processing data from 1,000+ regulated financial institutions.
- Improved reporting workflows for economic surveillance and monetary policy analytics, reducing reporting and data retrieval latency by 40% through query optimization and scalable data processing pipelines.
- Developed an enterprise semantic search and **Retrieval-Augmented Generation (RAG)** platform using OpenAI embeddings, **vector search**, advanced chunking strategies, and metadata filtering across **100K+ internal documents**, improving retrieval relevance by **35%** and reducing search latency by 40%.
- Led migration of legacy Talend ETL workflows to scalable **PySpark/PySQL pipelines in Databricks**, implementing reusable ingestion, validation, CDC, and SCD1/SCD2 frameworks.
- Built end-to-end analytics solutions using **SQL, Tableau, and Power BI**, enabling self-serve insights and improving business KPIs by **8-12%**.
- Developed responsive web dashboards and reporting portals using **React.js** and **REST APIs**, reducing manual reporting effort by 35% and improving data accessibility across cross-functional teams.
- Scaled distributed **ETL pipelines** using Python, Talend, and Spark to process **~2M rows/day**; optimized partitioning, caching, and I/O operations to reduce runtime by 20% and infrastructure costs by 16%.

Skills

Languages: Python, Java, Go, C++, SQL, Shell scripting, JavaScript, TypeScript

Frameworks: FastAPI, Spring Boot, React.js, Next.js, Node.js, Flask, gRPC

Backend & Architecture: RESTful APIs, Microservices, Distributed Systems, OAuth2, JWT, WebSockets, WebRTC, Agile/Scrum

Cloud & DevOps: AWS (EC2, ECS, EKS, Lambda, S3, SageMaker), GCP, Azure, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Kafka, PySpark, Databricks

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, LangGraph, RAG, FAISS, MCP, Vector Search, Semantic Search, Embeddings, Fine-Tuning, CUDA, Prompt Engineering, Guardrails, LLM, AI Agents, Multi-Agent Systems, AI Orchestration, Model Routing, Sentence Transformers

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase

Testing: pytest, JUnit, Unit Testing, Integration Testing

Developer Tools: Git, Linux, Docker, JIRA, Postman, Prometheus, Grafana, Tableau, Power BI, VS Code, Google Colab

Education

California State University, Chico - **Master of Science in Computer Science**

Aug 2024 – May 2026

University of Mumbai - **Bachelor of Engineering in Computer Engineering**

Aug 2017 – May 2020

Technical Projects

Multi-Agent Code Intelligence CLI (CoderAI)

- Built an agentic multi-LLM CLI with dynamic model routing, hierarchical AI agents, tool calling via MCP, and sandboxed code execution, enabling automated debugging, code generation, and web-assisted reasoning workflows.

Fine-Tuning Large Language Models - Gemma 4 E2B

- Fine-tuned Gemma 4 E2B via OOP-structured supervised instruction tuning pipeline with LoRA (90% fewer trainable parameters) and 4-bit quantization, reducing model storage by 30% and enabling single consumer-GPU deployment

Semantic Search Engine - RAG Pipeline

- Built a Retrieval-Augmented Generation (RAG) pipeline with web ingestion, Sentence Transformer embeddings, FAISS indexing, and cosine-similarity re-ranking, improving Top 3 retrieval precision over baseline vector retrieval.

TurboQuant - KV Cache Quantization for LLM Inference

- Benchmarked KV-cache quantization for transformer inference (FP16 → INT4), achieving 50% GPU memory reduction with <2% perplexity degradation, enabling longer context windows and more efficient LLM inference on constrained hardware.

Chess Engine

- Developed a chess engine using the Min-Max algorithm with Alpha-Beta pruning to efficiently predict optimal moves by reducing the search space. Used RESTful Flask API for backend development and JavaScript for creating interactive UI.

Hackathon

Wefunder AI Hackathon – Context Router (Winner)

- Developed Context Router, an intelligent LLM-routing system that dynamically analyzes user queries and selects the most suitable large language model LLM based on criteria like token length, task complexity, and required reasoning depth.
- Won the Pond Challenge** among 20+ teams; received recognition for designing a novel LLM orchestration strategy that reduced infrastructure waste and introduced a new paradigm for modular, cost-aware AI deployments.
- Achieved up to 20% reduction in **API costs** by implementing **prompt classification**, and **cost-aware model switching**.